
Applications for Laser Photodetachment for Ion Particle Accelerators

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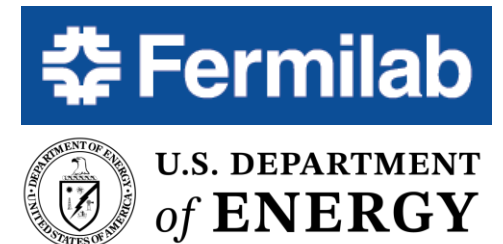
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Boston University Advisor: Professor Ed Kearns

Fermilab Supervisors: Jinhao Ruan & Randy Thurmman-Keup

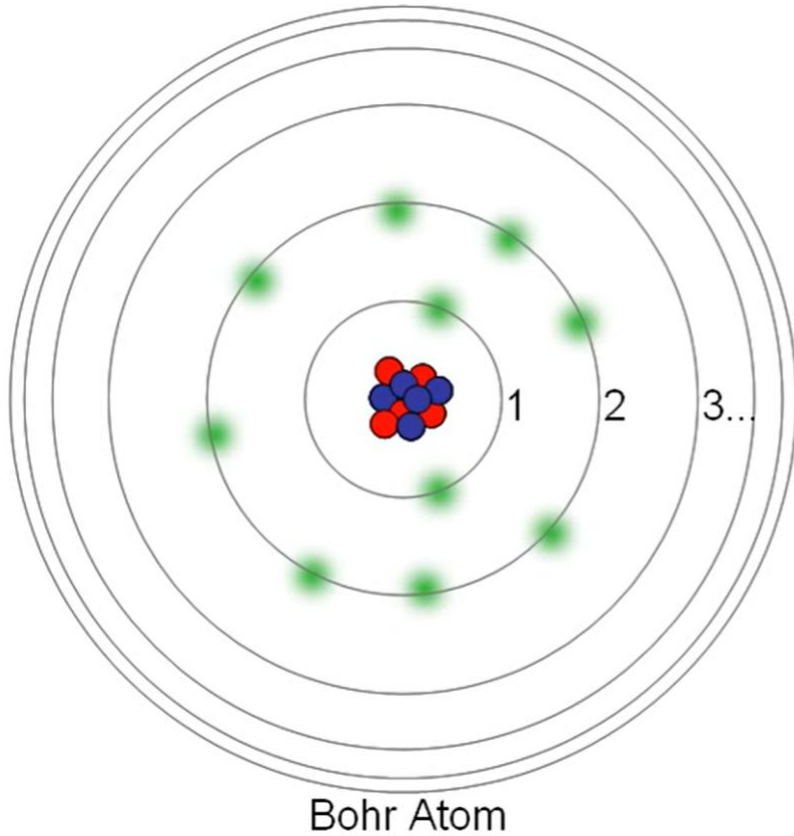
Collaborators: Erin Chen, Ralitsa Sharankova & Vic Scarpine



Overview

- QM Review
- Intro to Lasers
- Laser Ion Interaction
- Applications in Accelerators

Quantum Mechanics Refresher



Schrodinger Equation:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{\mathcal{H}} \psi$$

Hamiltonian:

$$H = T + V$$

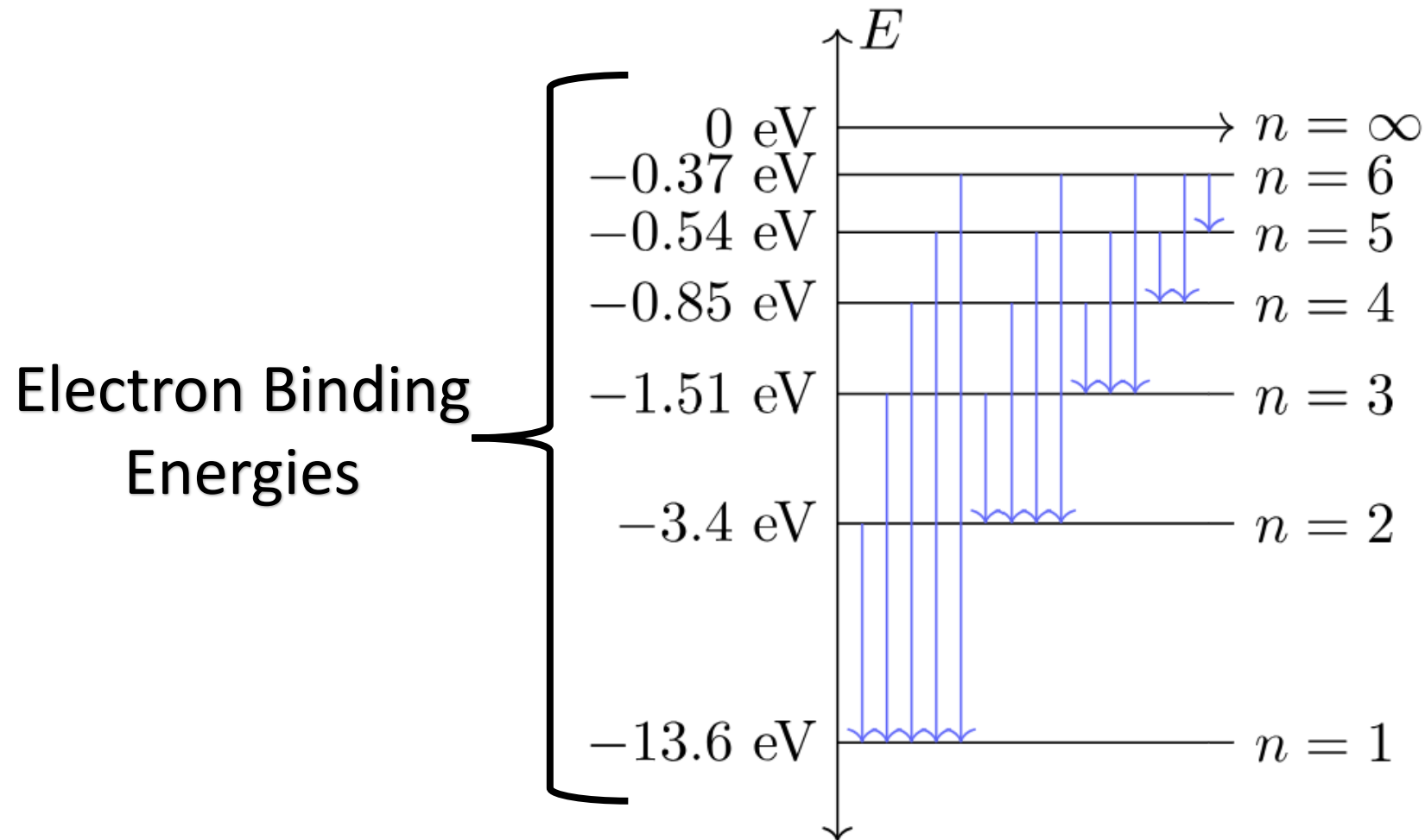
Kinetic Energy

Potential

Ridiculously Simplified Review of Quantum Mechanics- part I

<https://www.youtube.com/watch?v=yDq8XGveFrI&t=146s>

Quantum Mechanics Refresher



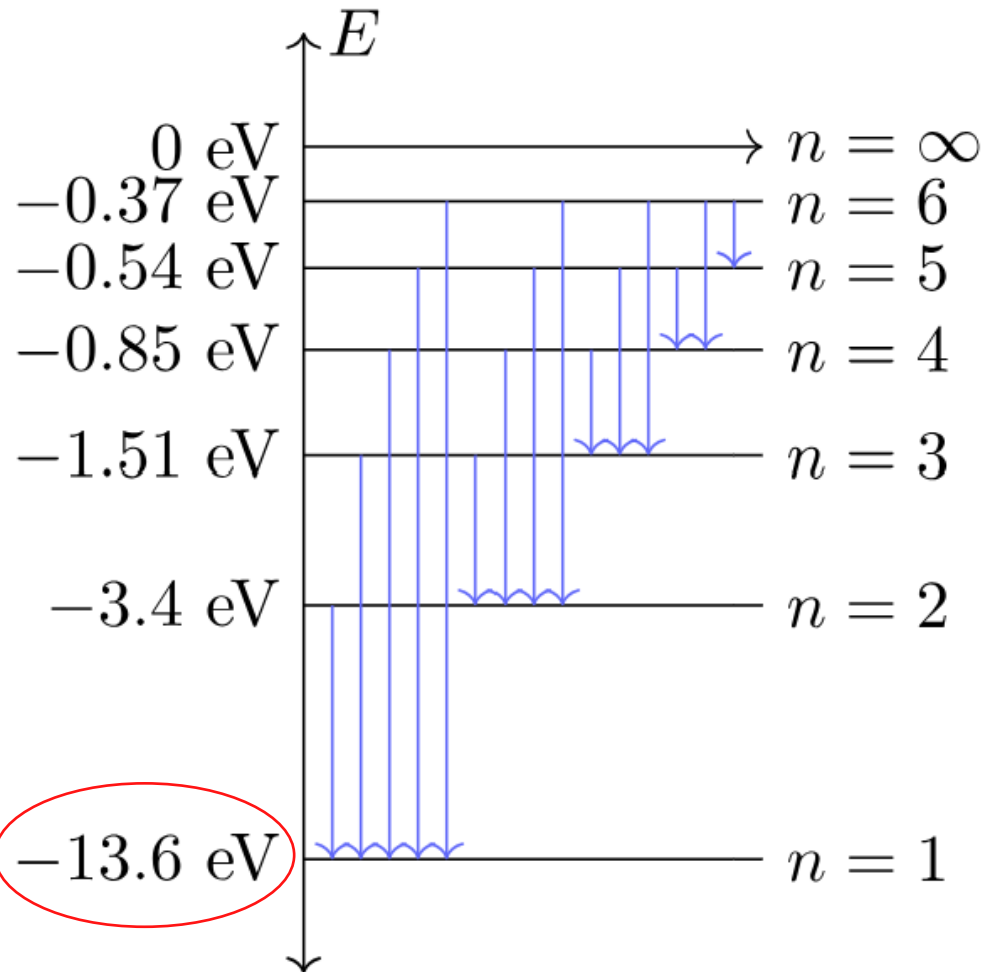
Hydrogen Energy Levels

<https://www.mit.edu/~ashrstnv/bohr-atom.html>

Quantum Mechanics Refresher

Derived from Bohr model
using Coulomb's Law:

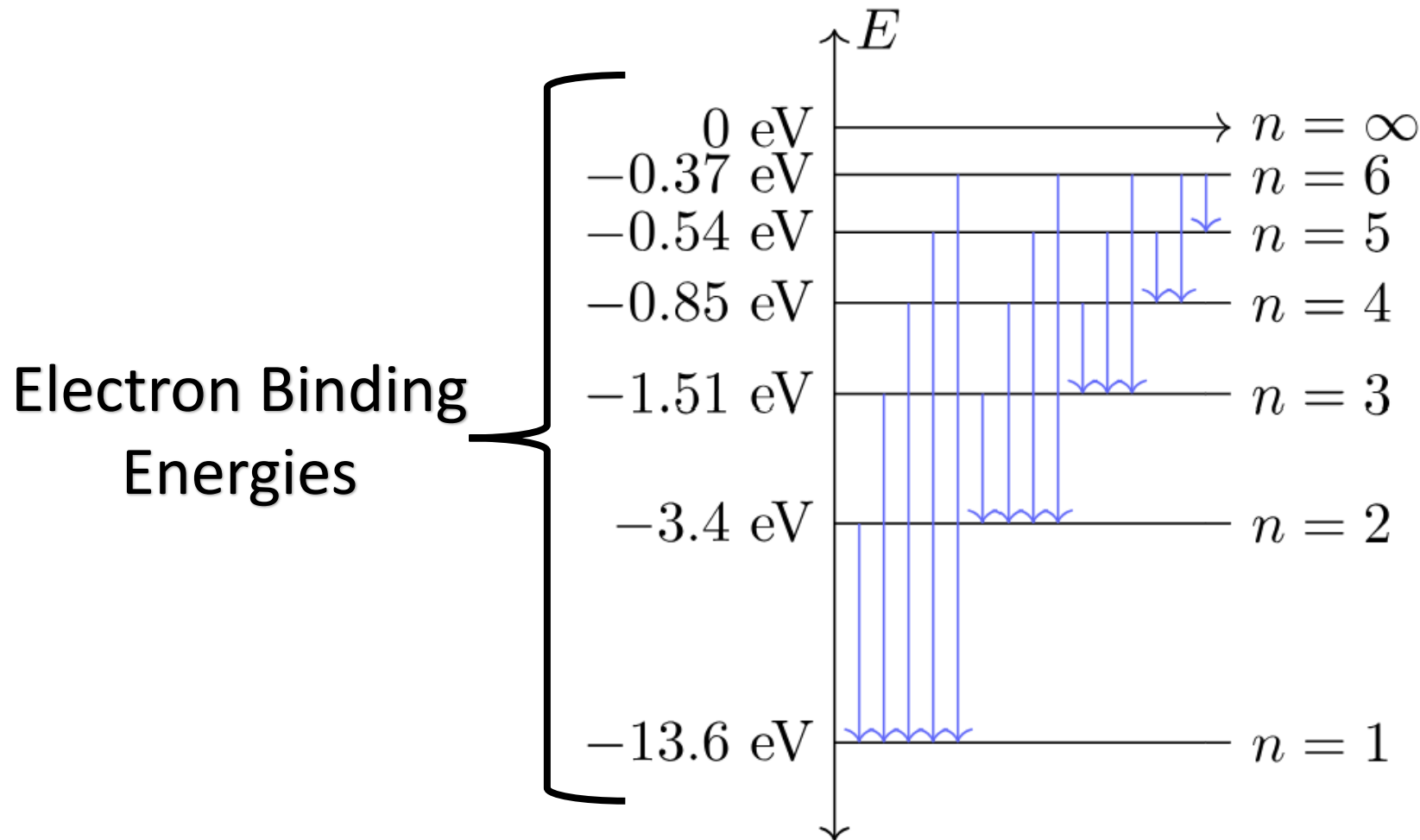
$$E_n = -\frac{m_e e^4}{32\pi^2 \epsilon_0^2 \hbar^2} \frac{1}{n^2}$$



Hydrogen Energy Levels

<https://www.mit.edu/~ashrstnv/bohr-atom.html>

Quantum Mechanics Refresher



Energy of a State:

$$E_n[\text{eV}] = -\frac{13.6}{n^2}$$

Transition Energy

$$\Delta E = E_{\text{final}} - E_{\text{initial}}$$

Hydrogen Energy Levels

<https://www.mit.edu/~ashrstnv/bohr-atom.html>

Quantum Mechanics Refresher

$$E_{\gamma} = \hbar \omega$$

Energy of photon

Reduced Planck constant

Angular frequency ($\omega = \frac{2\pi c}{\lambda_{\gamma}}$)

Energy of photon in eV

$$E_{\gamma} [eV] \approx \frac{1239.8}{\lambda [nm]}$$

Quantum Mechanics Practice Question

Radiative Decay:

An electron starting in state $n=4$ falls to $n=2$.

What is the λ_γ in nm that is emitted?

Quantum Mechanics Practice Question

Radiative Decay:

An electron starting in state $n=4$ falls to $n=2$.

What is the λ_γ in nm that is emitted?

$$\Delta E = E_f - E_i$$

$$\Delta E_{4 \rightarrow 2} = -\frac{13.6}{4^2} - \left(-\frac{13.6}{2^2} \right)$$

$$\Delta E_{4 \rightarrow 2} = 2.55 \text{ eV}$$

$$\lambda_\gamma = \frac{1239.8}{\Delta E_{4 \rightarrow 2}} = 486.20 \text{ nm}$$

Hydrogen Emission spectrum



What's a Laser?

LASER – Light **A**mplification by **S**timulated **E**mission of **R**adiation

3 elements that define a Laser

1. **Gain Medium**

Material with quantized energy levels

Examples: Nd:YAG crystal, He-Ne gas, semiconductor diode

2. **Pump (Excitation Source)**

Supplies energy to create population inversion

Electrical discharge, optical pumping

3. **Optical Resonator (Cavity)**

Mirrors provide optical feedback.

Cavity forces light to pass gain medium multiple times

What's a Laser?

Spontaneous Emission

Radiative Decay:

An electron starting in state $n=4$ falls to $n=2$.
What is the λ_γ in nm that is emitted?

**Our example
from earlier:**

$$\Delta E = E_f - E_i$$

$$\Delta E_{4 \rightarrow 2} = -\frac{13.6}{4^2} - \left(-\frac{13.6}{2^2}\right)$$

$$\Delta E_{4 \rightarrow 2} = 2.55 \text{ eV}$$

$$\lambda_\gamma = \frac{1239.8}{\Delta E_{4 \rightarrow 2}} = 486.20 \text{ nm}$$

Hydrogen Emission spectrum



**The emitted photon will be in a
random direction and phase!**

What's a Laser?

$\hat{H}$: Hamiltonian

E_n : Energy Levels

ψ_n : Wavefunctions of electron

Stimulated Emission

$$\hat{H}\psi_n = E_n\psi_n$$

What's a Laser?

$\hat{H}$: Hamiltonian

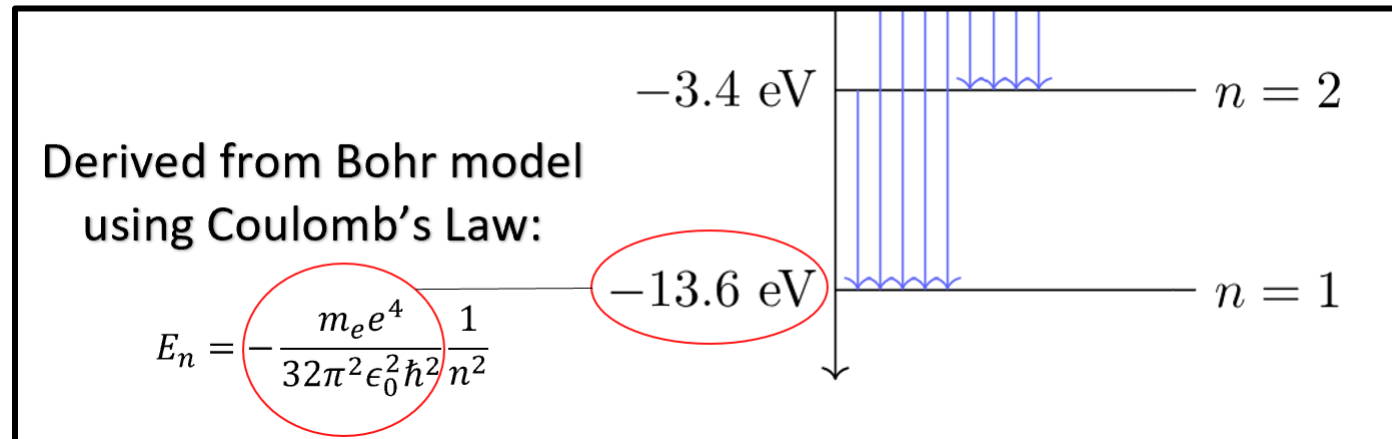
E_n : Energy Levels

ψ_n : Wavefunctions of electron

Stimulated Emission

$$\hat{H}\psi_n = E_n\psi_n$$

$$\hat{H} \rightarrow E_n$$



$$\hat{\mathcal{H}} = -\frac{\hbar}{2m_e} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r}$$

What's a Laser?

$\hat{H}$: Hamiltonian

E_n : Energy Levels

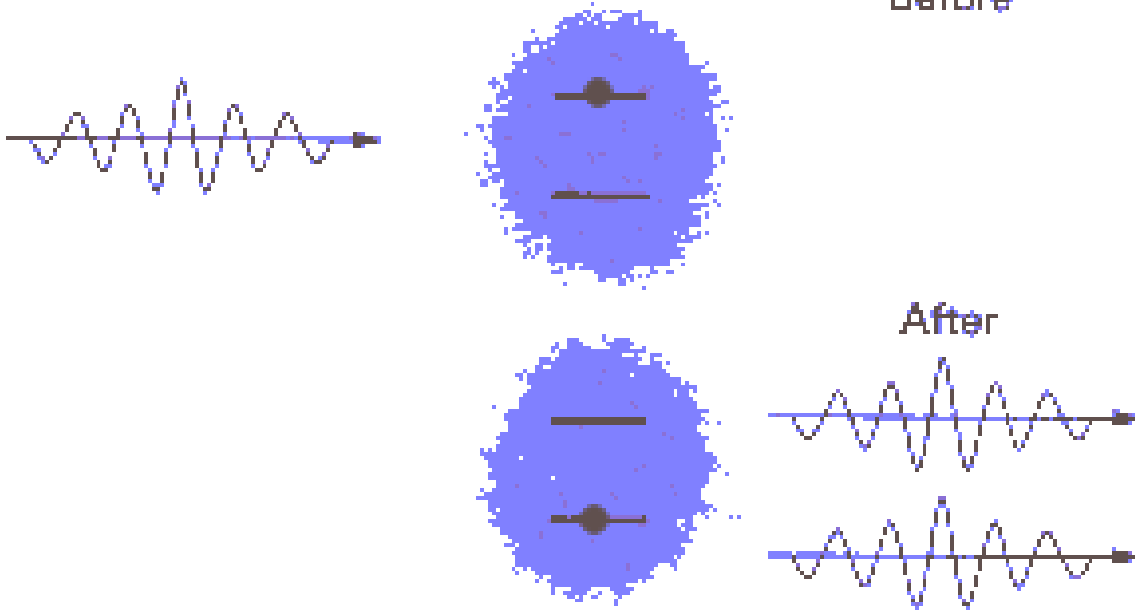
ψ_n : Wavefunctions of electron

Stimulated Emission

$$\hat{H}\psi_n = E_n\psi_n$$

$$\hat{\mathcal{H}}_{stim} = H_0 + H_{laser}$$

STIMULATED EMISSION



What's a Laser?

$\hat{H}$: Hamiltonian

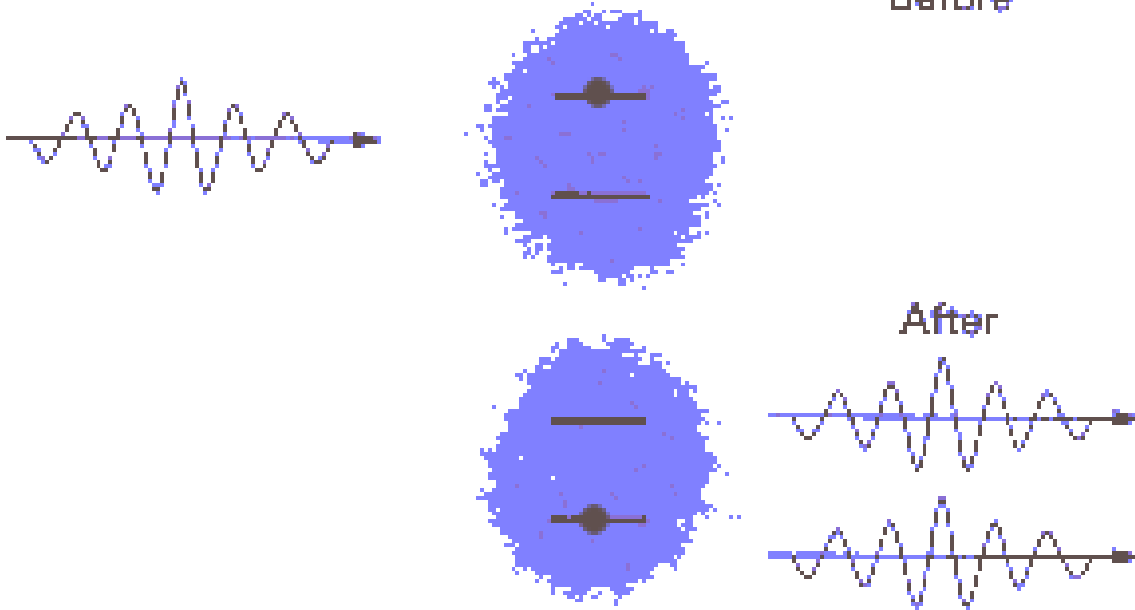
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STIMULATED EMISSION



$$\widehat{\mathcal{H}}_{stim} = H_0 + H_{laser}$$

Laser E-field

$$E(t) = E \cos(\omega t)$$

Dipole Potential

$$H_{laser} = -d \cdot E(t)$$

What's a Laser?

$\hat{H}$: Hamiltonian

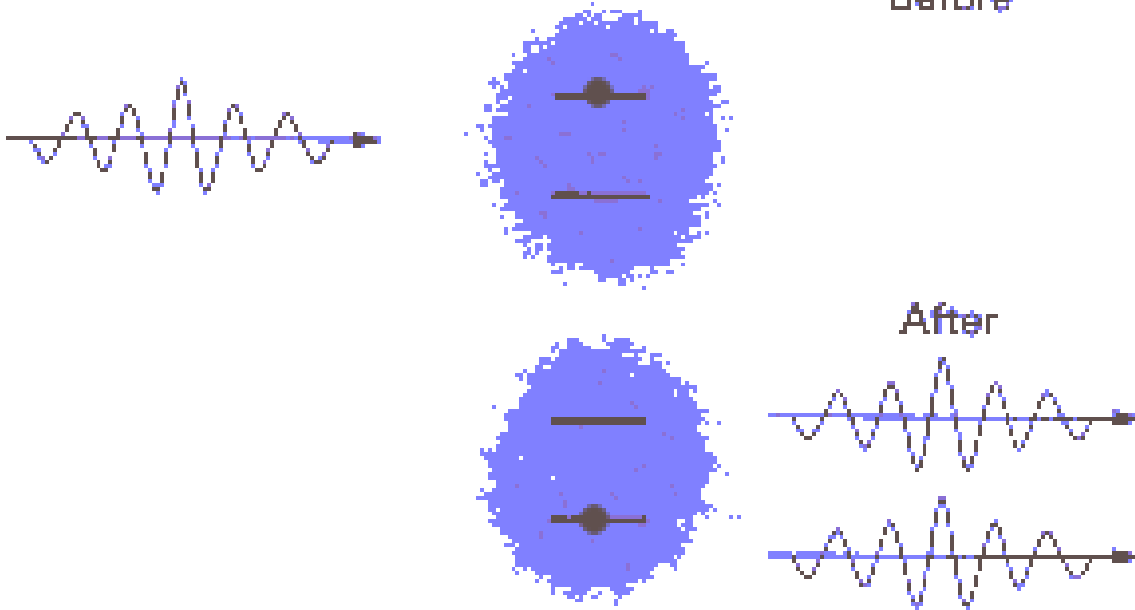
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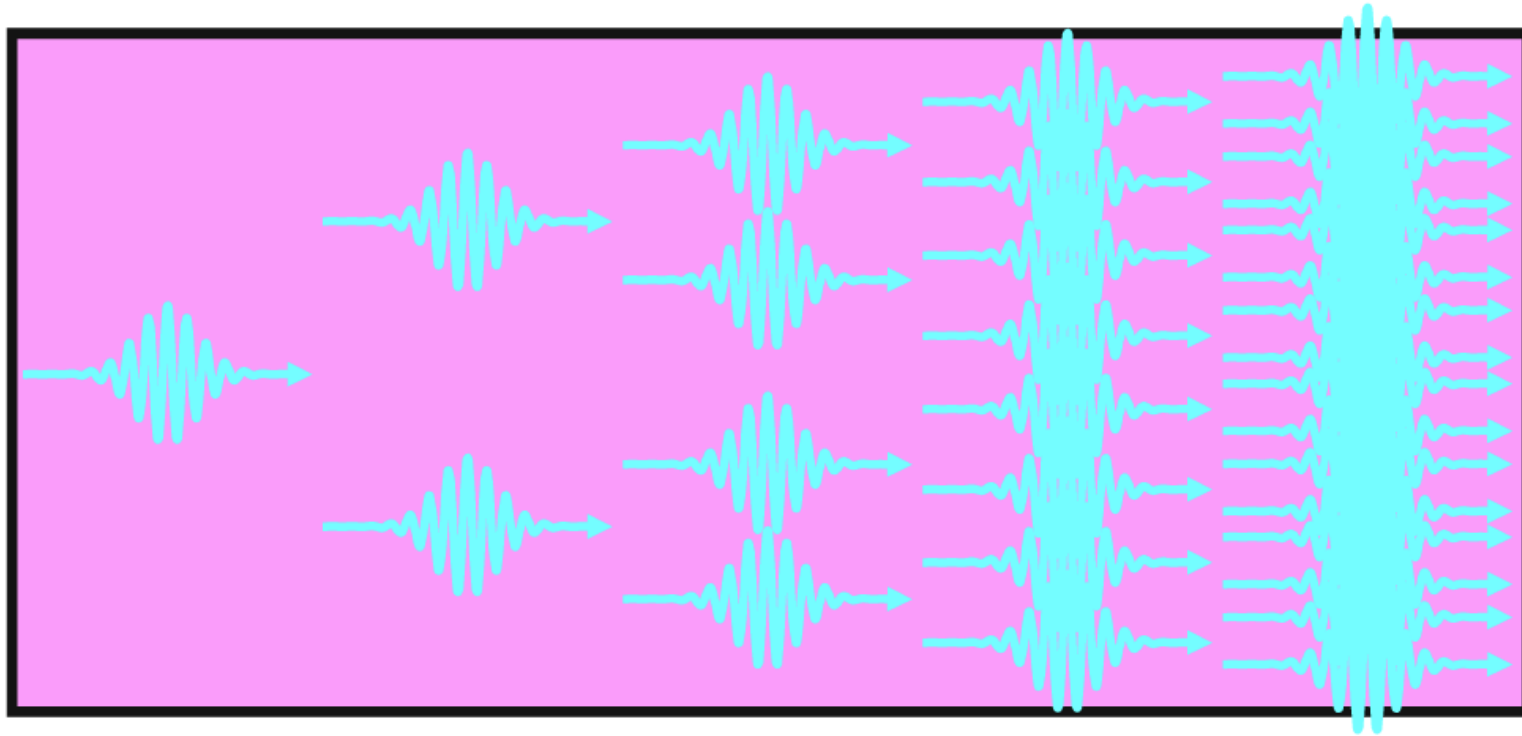
Dipole Potential

$$H_{laser} = -d \cdot E(t)$$

Electron is moved back and forth in laser field, so their phases match!

That's a Laser!

Excited medium



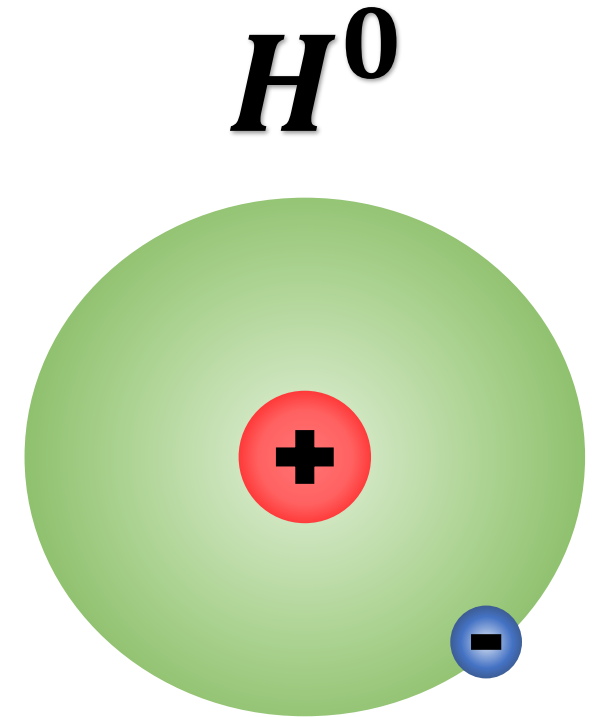
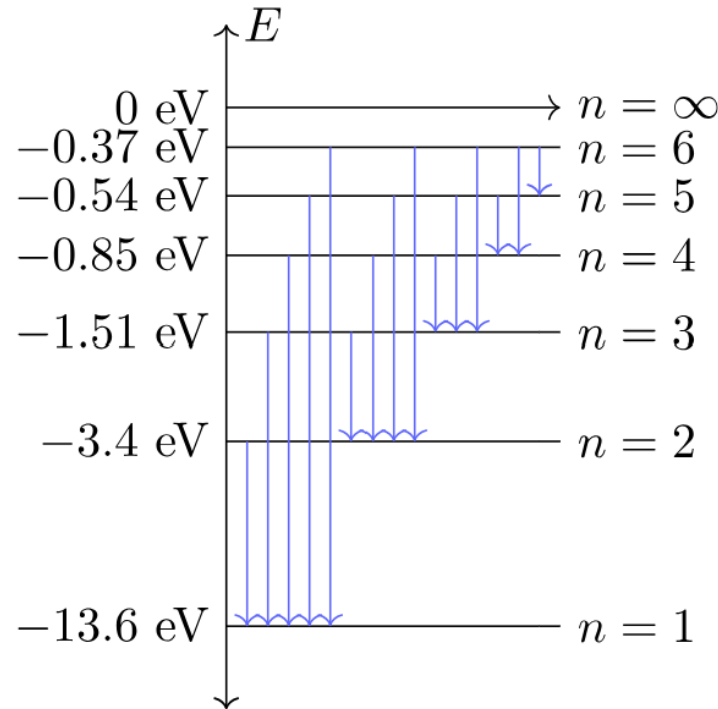
- We reviewed QM hydrogen atom.
- Applied these concepts to get coherent emissions.
- Now why would this be important?

Back to Particles: Now what happens if we go back to Ions

Bohr Model assumes that electron only “feels” the proton

$$E_n = - \underbrace{\frac{m_e e^4}{32\pi^2 \epsilon_0^2 \hbar^2}}_{e^2} \frac{1}{n^2}$$

$$\widehat{\mathcal{H}} = -\frac{\hbar}{2m_e} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r}$$



Now what happens if we change to Heavier Ions

$$H = T + V$$

$$H_{H^-} = T + V_{eN} + V_{ee}$$

$$H_{Heavy} = \sum (T_i + V_{iN}) + \sum (V_{ij})$$

Kinetic energy and potential
with nucleus

Electron/Electron
potential

Now what happens if we change to Heavier Ions

$$H = T + V$$

$$H_{H^-} = T + V_{eN} + V_{ee}$$

$$H_{Heavy} = \sum (T_i + V_{iN}) + \sum (V_{ij})$$

Momenta and potential
with nucleus

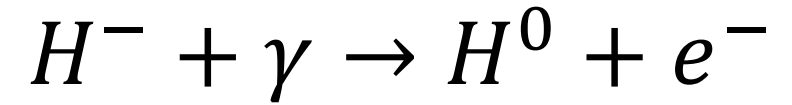
Electron/Electron
potential

Laser Driven Potential

$$- d \cdot E(t)$$

1. What wavelength?
2. What Intensity?

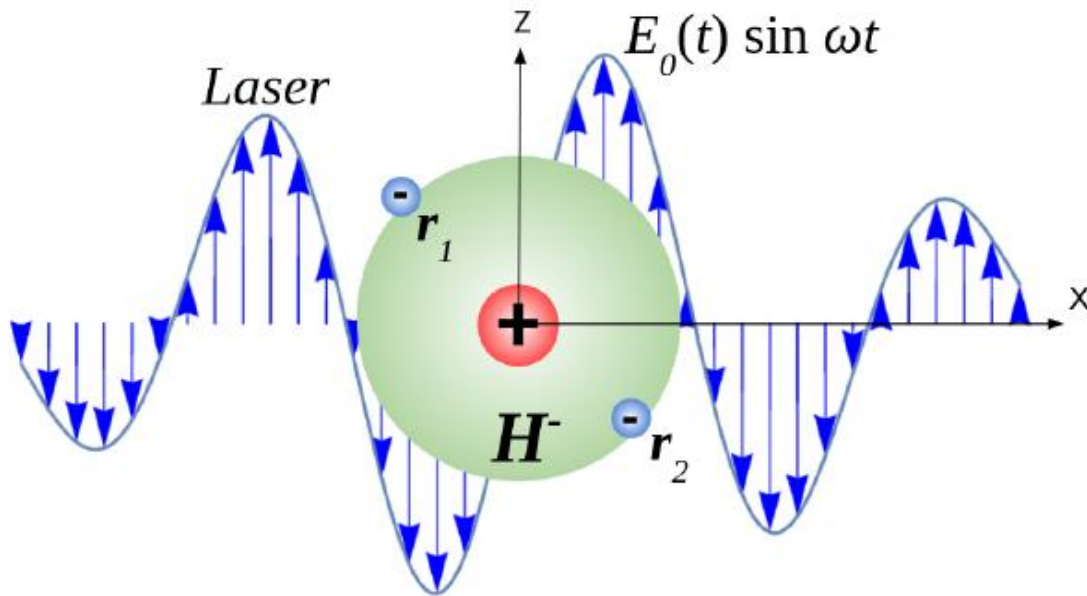
Laser Ion Interaction



Binding Energy of outer electron:

$$E_{e^{-}} = 0.754 \text{ eV}$$

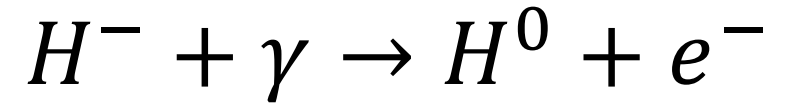
$$\lambda \cong 1645 \text{ nm}$$



PHOTODETACHMENT OF NEGATIVE HYDROGEN ION BEAM

<https://inspirehep.net/literature/2869019>

Laser Ion Interaction

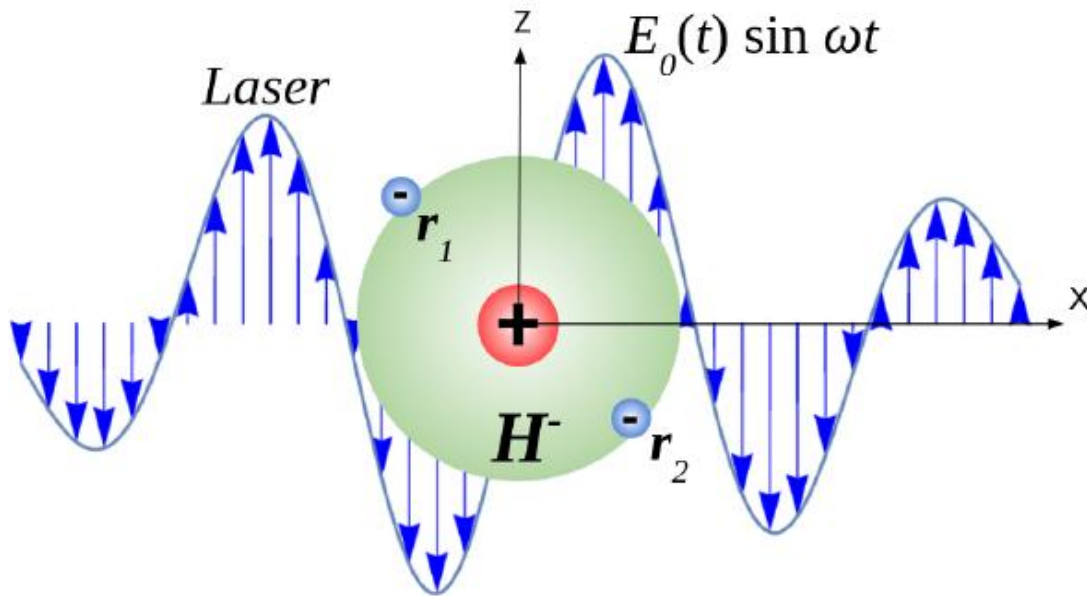


Binding Energy of outer electron:

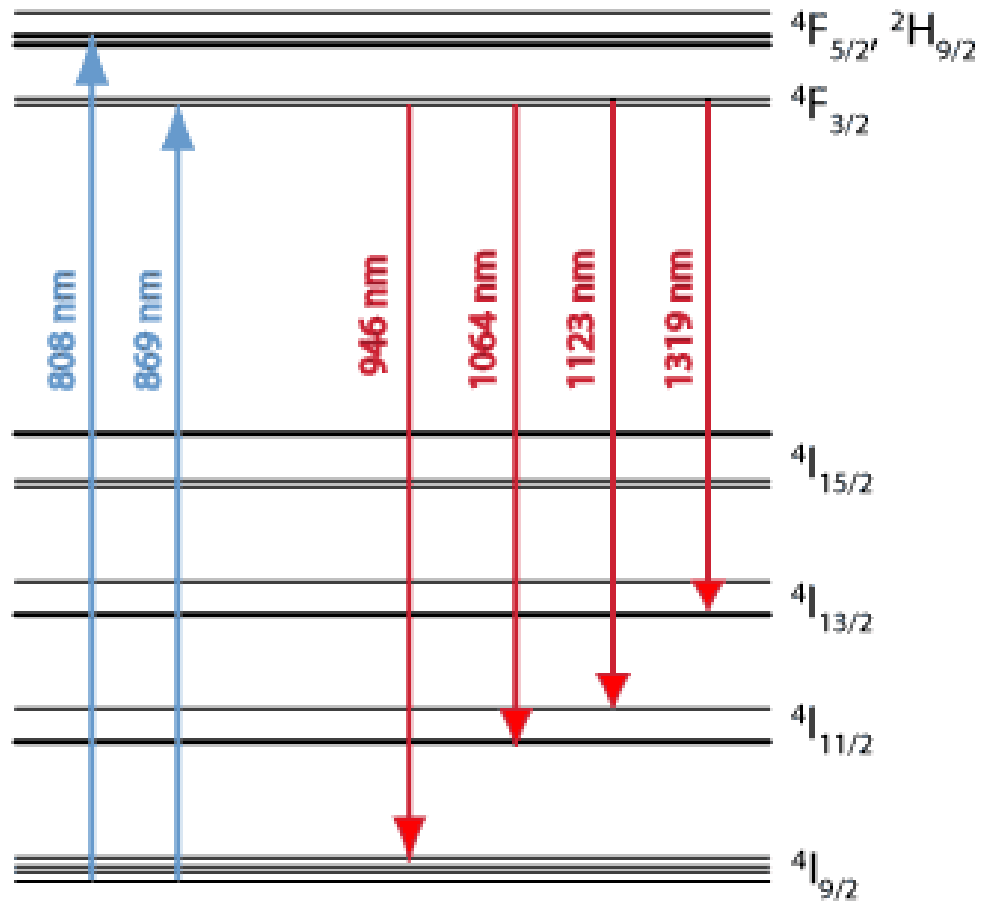
$$E_{e^{-}} = 0.754 \text{ eV}$$

$$\lambda \cong 1645 \text{ nm}$$

We can choose any laser with
a wavelength smaller than this!



Laser Ion Interaction

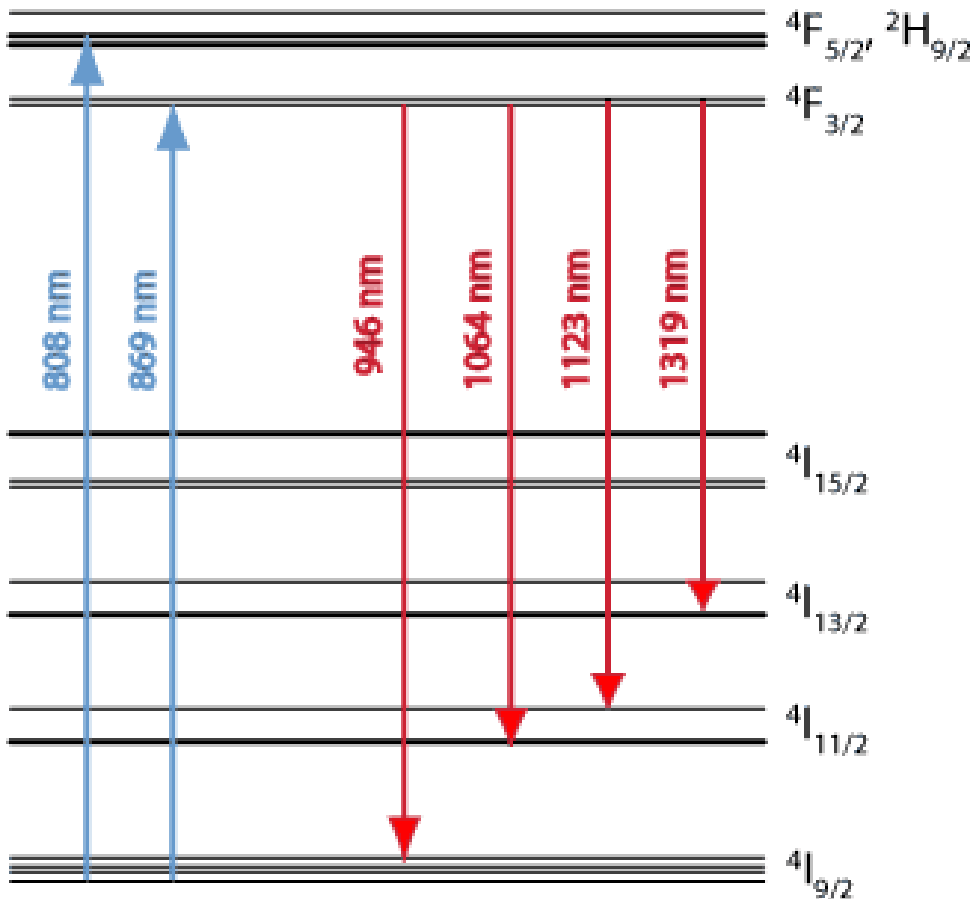


Our favorite
Rare-Earth-Doped crystal!

Laser Ion Interaction

Nd:YAG

Neodymium-doped Yttrium Aluminum Garnet



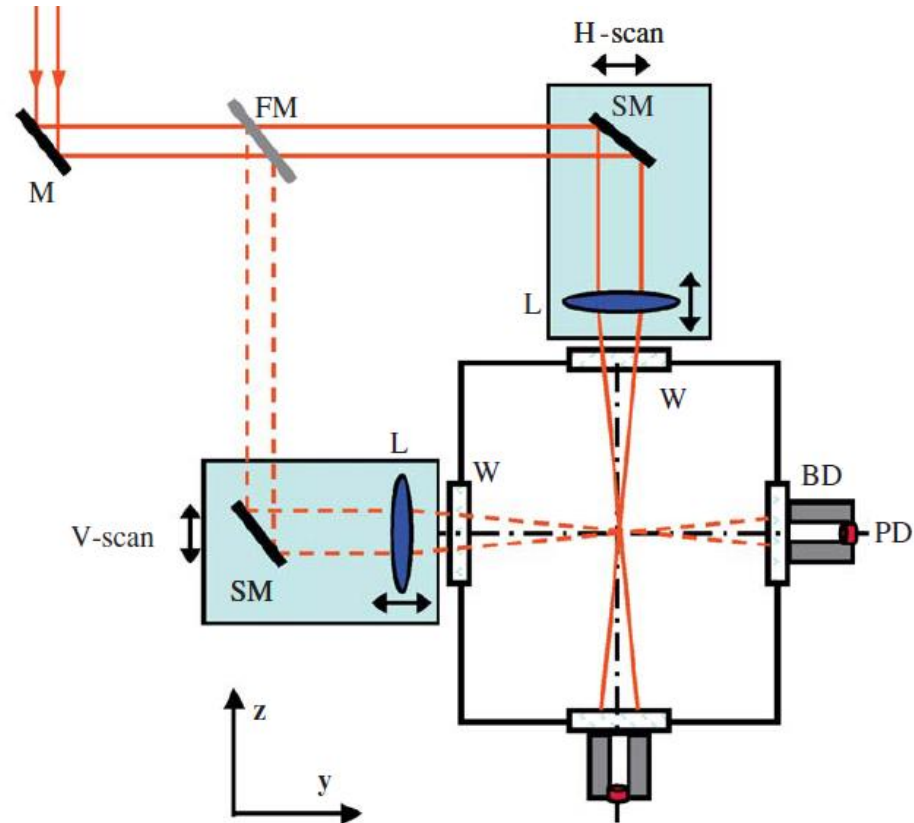
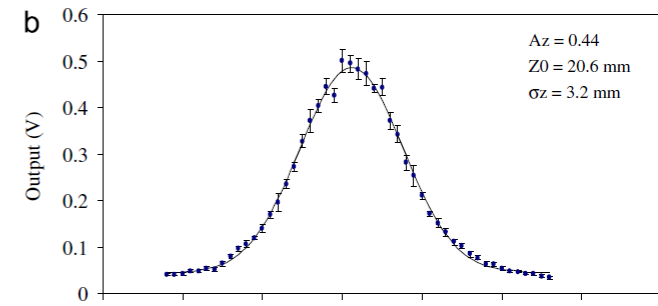
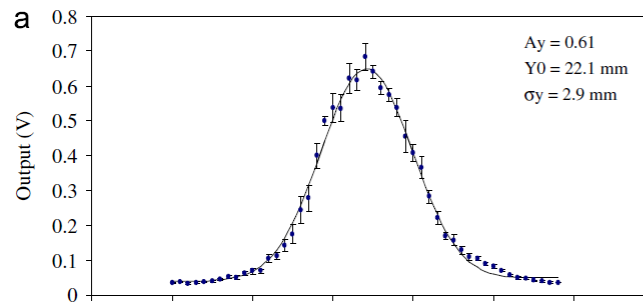
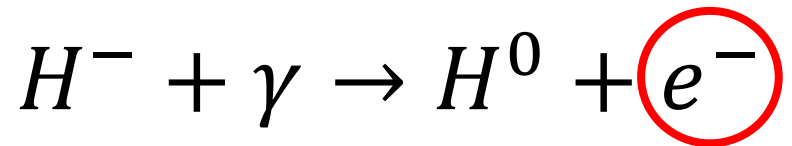
- We pump with $\sim 800\text{nm}$
- Doping decides which wavelength is produced.
- Fermilab, SNS, JPARC and more use 1064nm for instrumentation and beam diagnostics.

- We know a type of laser that works for H-interactions!
- Let us look at some applications...

SNS Laser Wire

[Video](#)

[Demonstration](#)



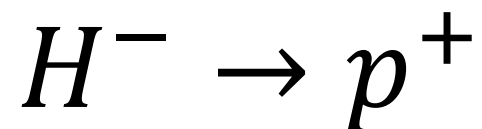
Recent progress in laser wire-based H^- beam diagnostics at the SNS linac

<https://proceedings.jacow.org/ipac2024/doi/jacow-ipac2024-tups85/>

Laser wire beam profile monitor in the spallation neutron source (SNS) superconducting linac:

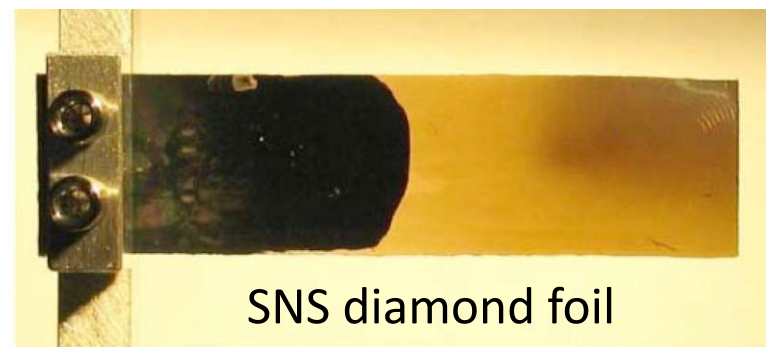
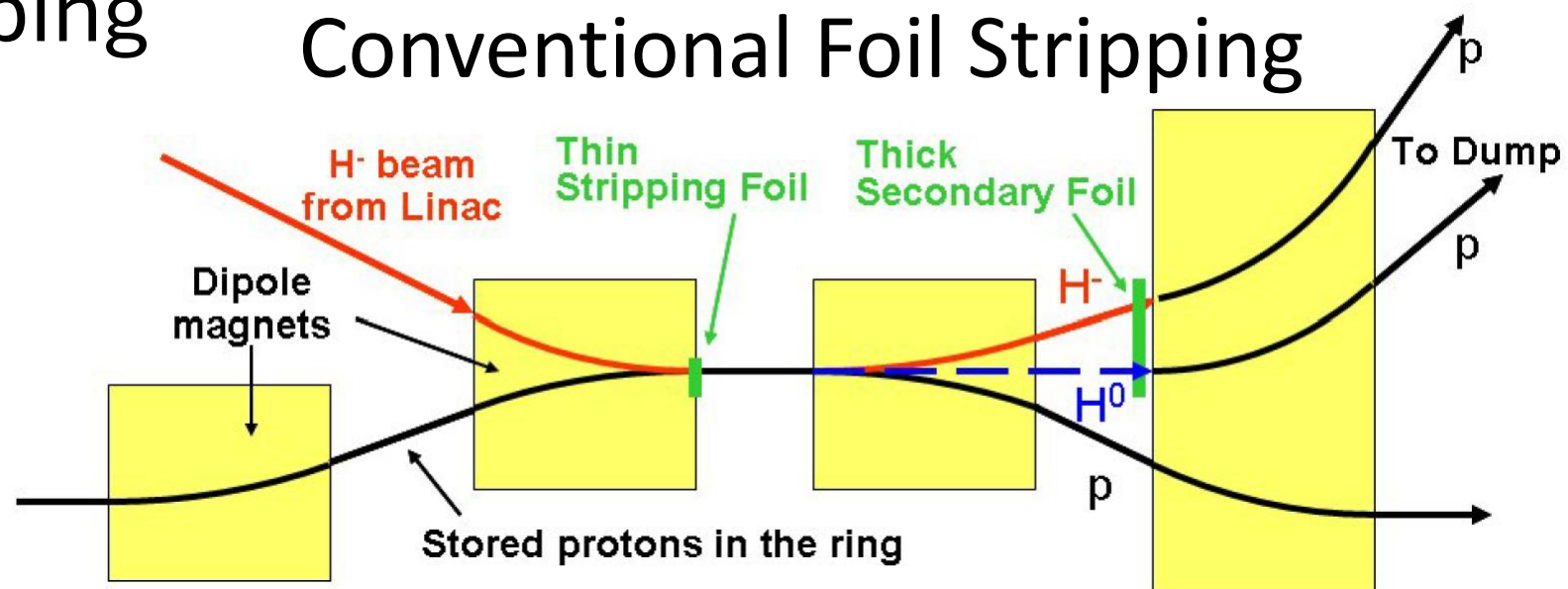
<https://www.sciencedirect.com/science/article/pii/S0168900209019858>

SNS & JPARC H- Stripping



- Foil degrades overtime
- Temperature Limits
- Scattering causes the beam to size increase

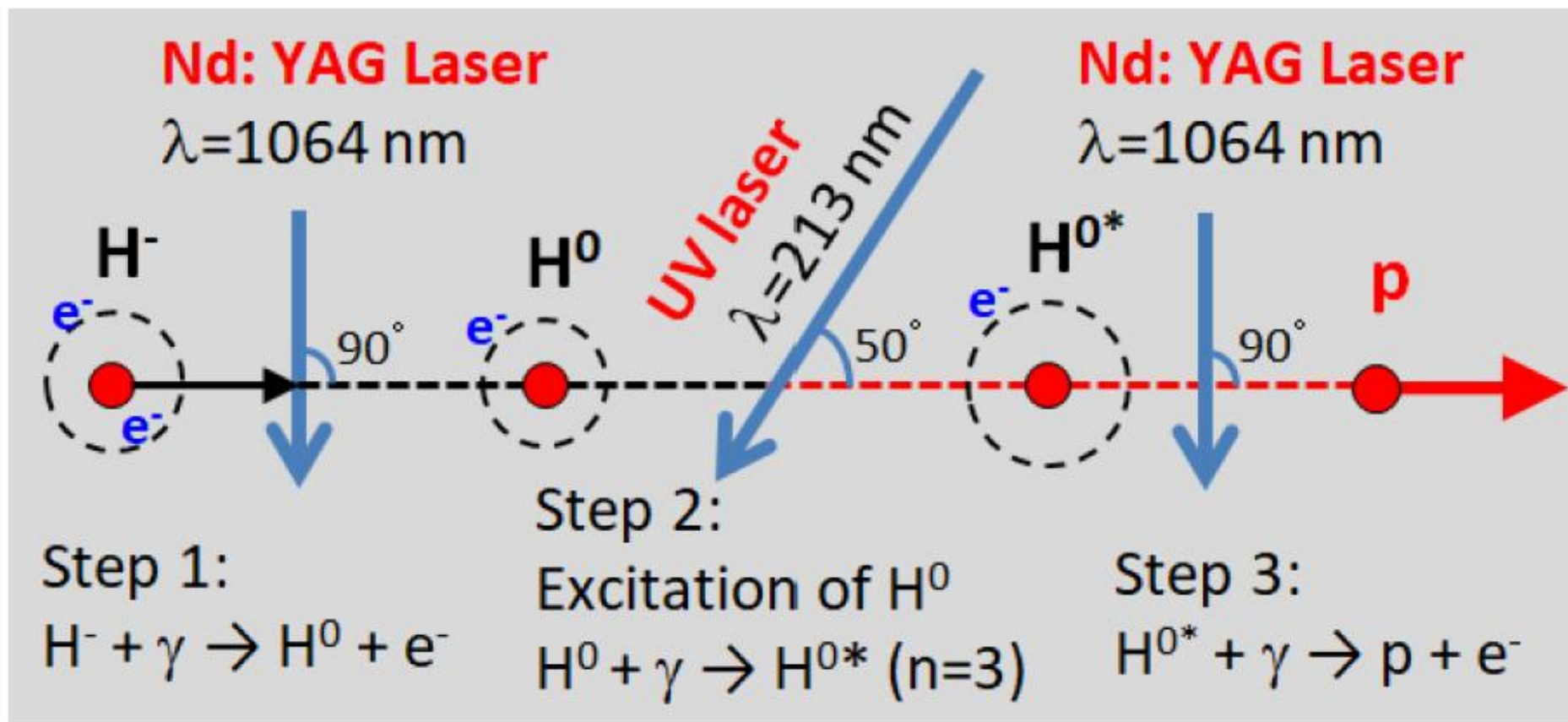
Conventional Foil Stripping



HIGH EFFICIENCY LASER-ASSISTED H- BEAM CONVERSION TO PROTONS

Masters Thesis of David Brown UI

JPARC Laser H- Stripping



5-YEAR PROGRESS OF H- LASER STRIPPING AT J-PARC RCS

Proceedings of the 21st Annual Meeting of Particle Accelerator Society of Japan

JPARC Laser H- Stripping

Why not do in one step?

- One Step Stripping: $E_{1 \rightarrow \infty} = 13.6 \text{ eV}$

JPARC Laser H- Stripping

Why not do in one step?

- One Step Stripping: $E_{1 \rightarrow \infty} = 13.6 \text{ eV} \rightarrow \lambda = 91.2 \text{ nm}$ (Deep UV)

Short answer, we would need an FEL

JPARC Laser H- Stripping

- First laser interaction we understand: $E_{e^-} = 0.754 \text{ eV}$
- Second interaction takes $E_1 \rightarrow E_3$

$$\Delta E = E_f - E_i$$

$$\Delta E_{1 \rightarrow 3} = -\frac{13.6}{3^2} - \left(-\frac{13.6}{1^2} \right)$$

$$|\Delta E_{1 \rightarrow 3}| = 12.08 \text{ eV}$$

$$\lambda_{1 \rightarrow 3} = \mathbf{102.6 \text{ nm}}$$

JPARC Laser H- Stripping

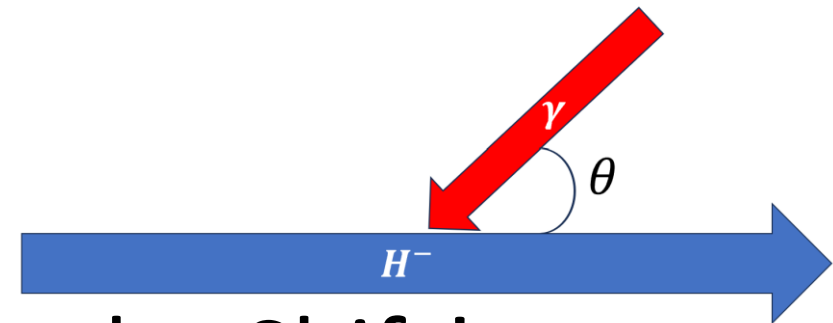
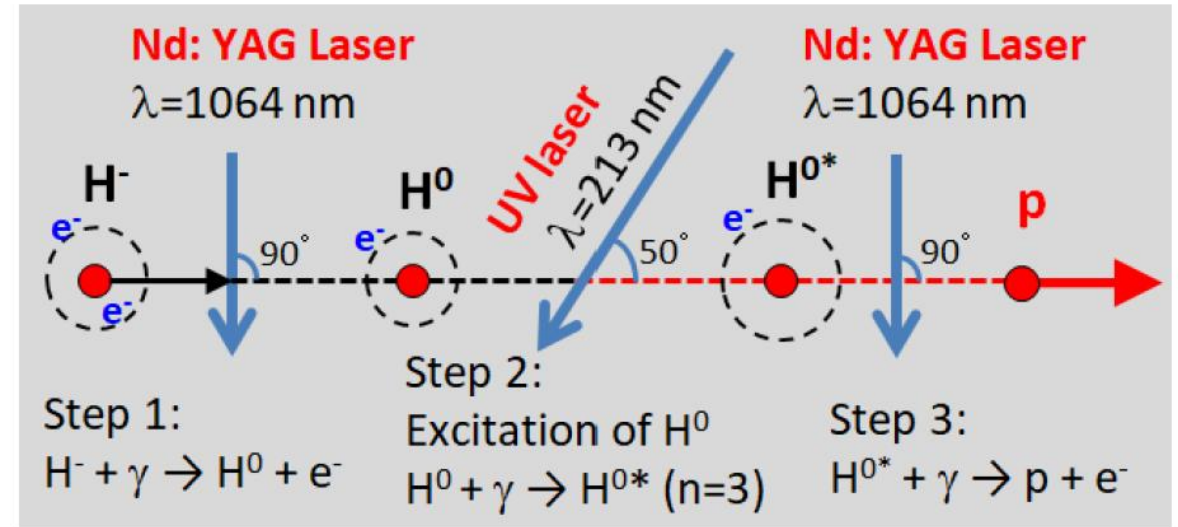
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Relativistic Doppler Shift!

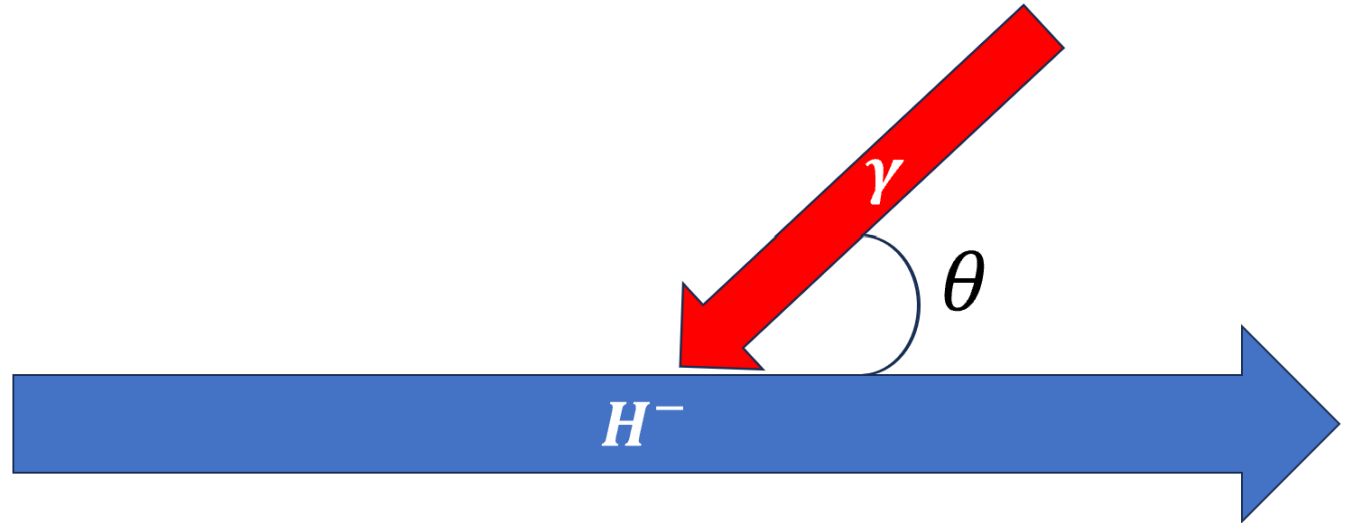
JPARC Laser H- Stripping

- First laser interaction we understand: $E_{e^-} = 0.754 \text{ eV}$
- Second interaction takes $E_1 \rightarrow E_3$

$$E_{H^-} = 400 \text{ MeV}$$

$$\theta = 50^\circ$$

$$\lambda_{start} = 213 \text{ nm}$$



JPARC Laser H- Stripping

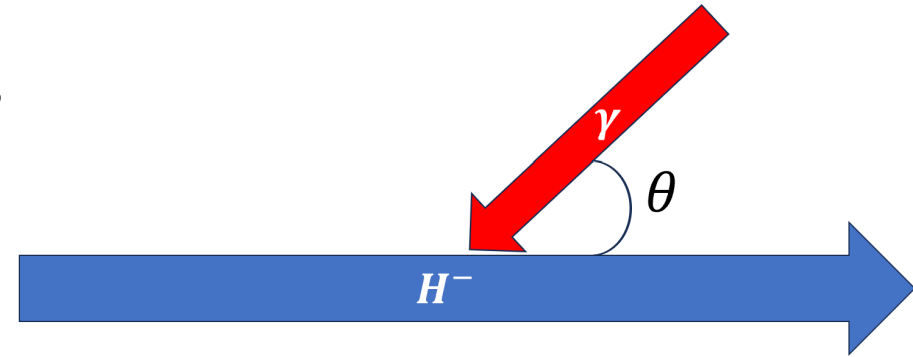
- First laser interaction we understand: $E_{e^-} = 0.754 \text{ eV}$
- Second interaction takes $E_1 \rightarrow E_3$

$$E_{H^-} = 400 \text{ MeV}$$

$$\theta = 50^\circ$$

$$\lambda_{start} = 213 \text{ nm}$$

$$\lambda_f = \frac{\lambda_{start}}{\gamma(1 + \beta \cos(\theta))}$$



$$\lambda_f = \frac{213}{\gamma(1 + \beta \cos(50^\circ))}$$

$$\lambda_f = 102.4 \text{ nm}$$

$$\lambda_{1 \rightarrow 3} = 102.6 \text{ nm}$$

JPARC Laser H- Stripping

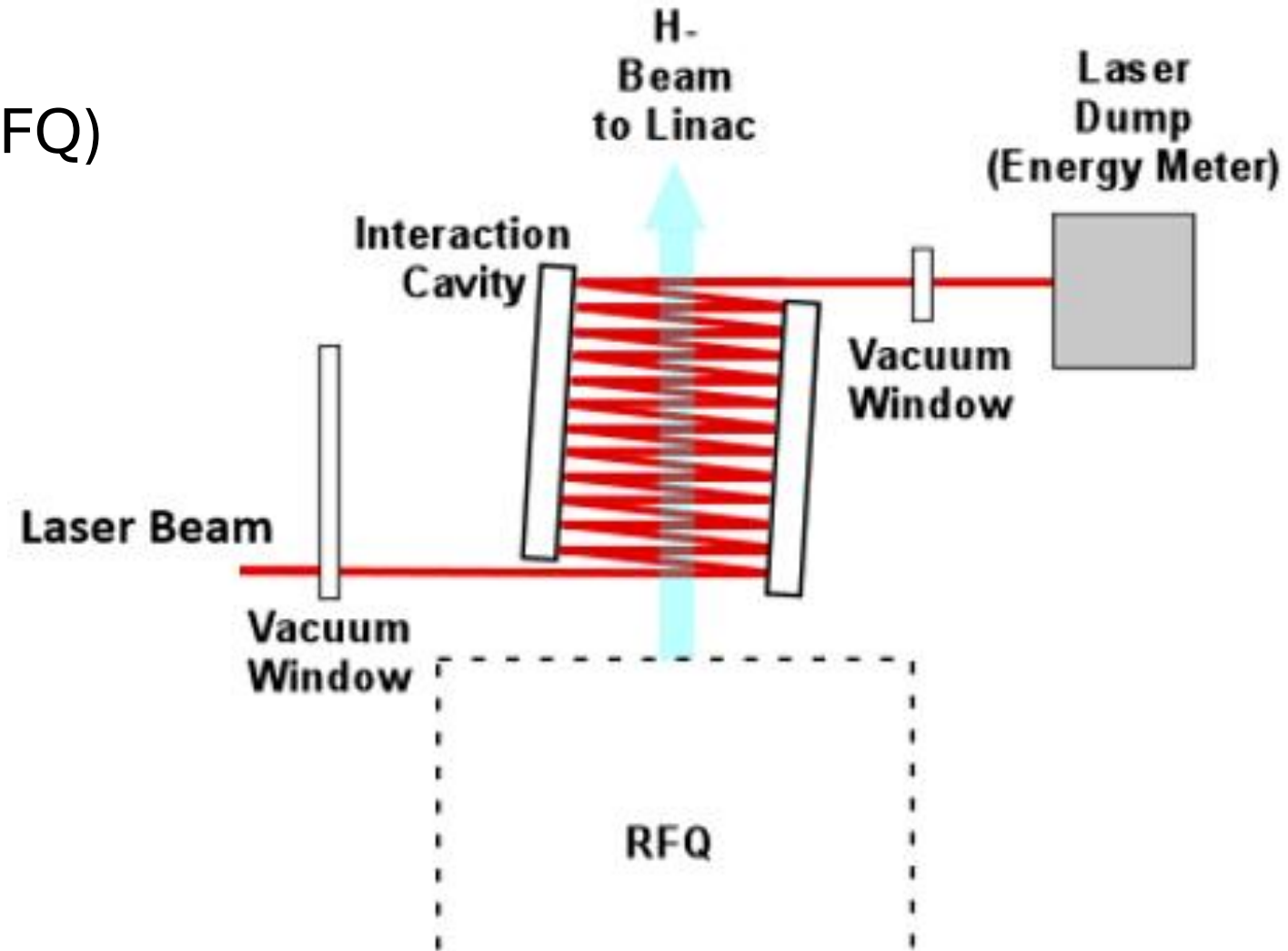
- First laser interaction we understand: $E_{e^-} = 0.754 \text{ eV}$
- Second interaction: $|\Delta E_{1 \rightarrow 3}| = 12.8 \text{ eV}$ where we doppler shift 213nm
- Lastly $|\Delta E_{3 \rightarrow \infty}| = 13.6 \text{ eV} - 12.8 \text{ eV} = 0.8 \text{ eV}$, $\lambda = 1549.8 \text{ nm}$
 - We can just use the same laser from the start!

Fermilab Laser H- Scraping (My work)

- Instrumentation
- H- Stripping
- Bunch-by-bunch Manipulation

Introducing the Laser Notcher

- Located right after the Radiofrequency Quadrupole (RFQ)
 $E_{H^-} = 750\text{keV}, @201.25\text{Mhz}$
Longitudinal Length: $\sim 2\text{ns}$

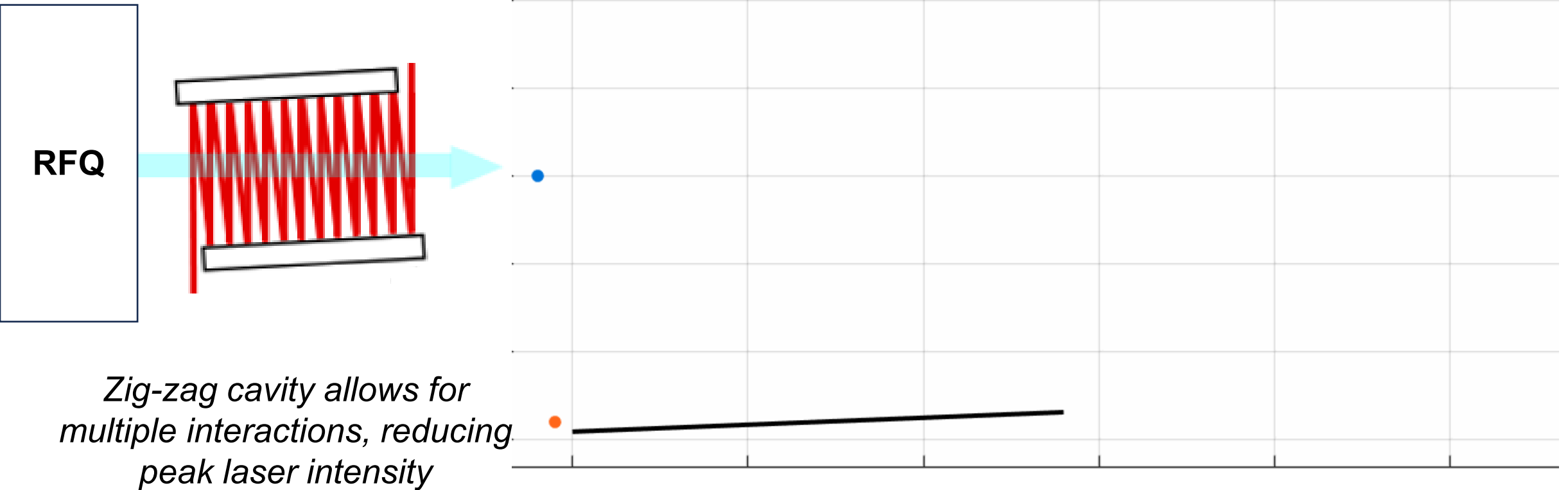


Introducing the Laser Notcher

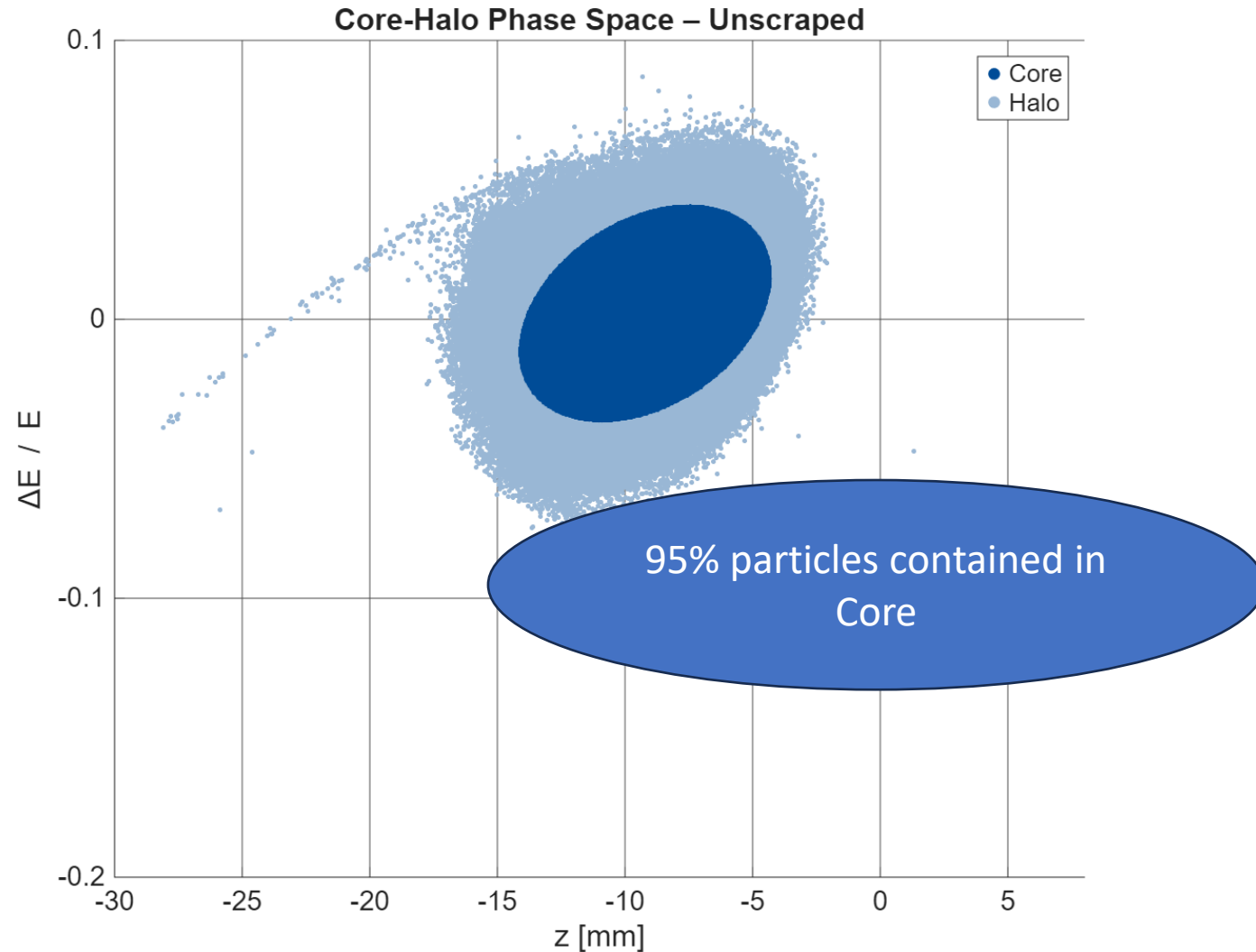
Optimal Following Angle:

$$\Theta = \arcsin(\beta)$$

Top-Down View

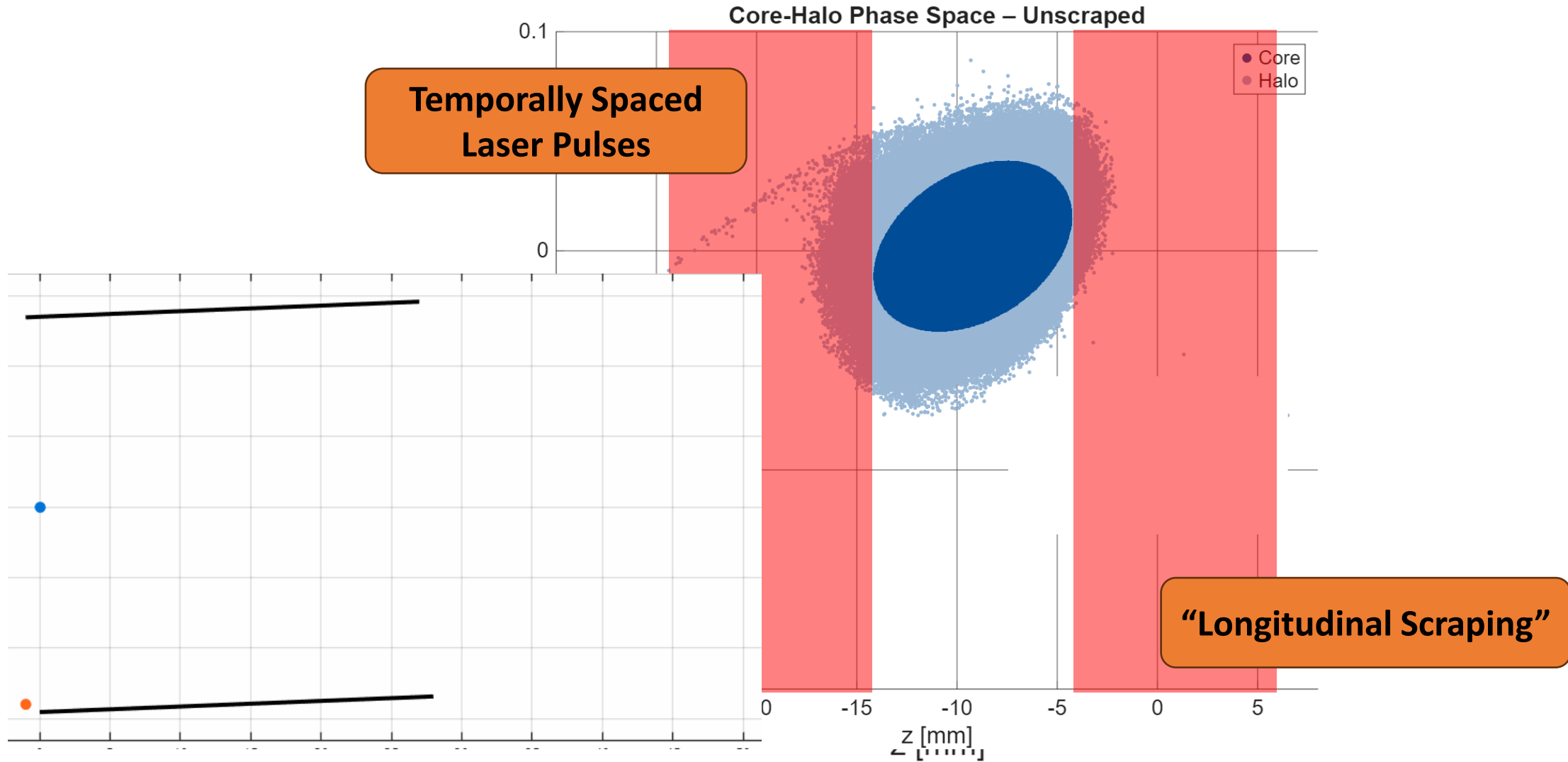


Fermilab Laser H- Scraping (My work)

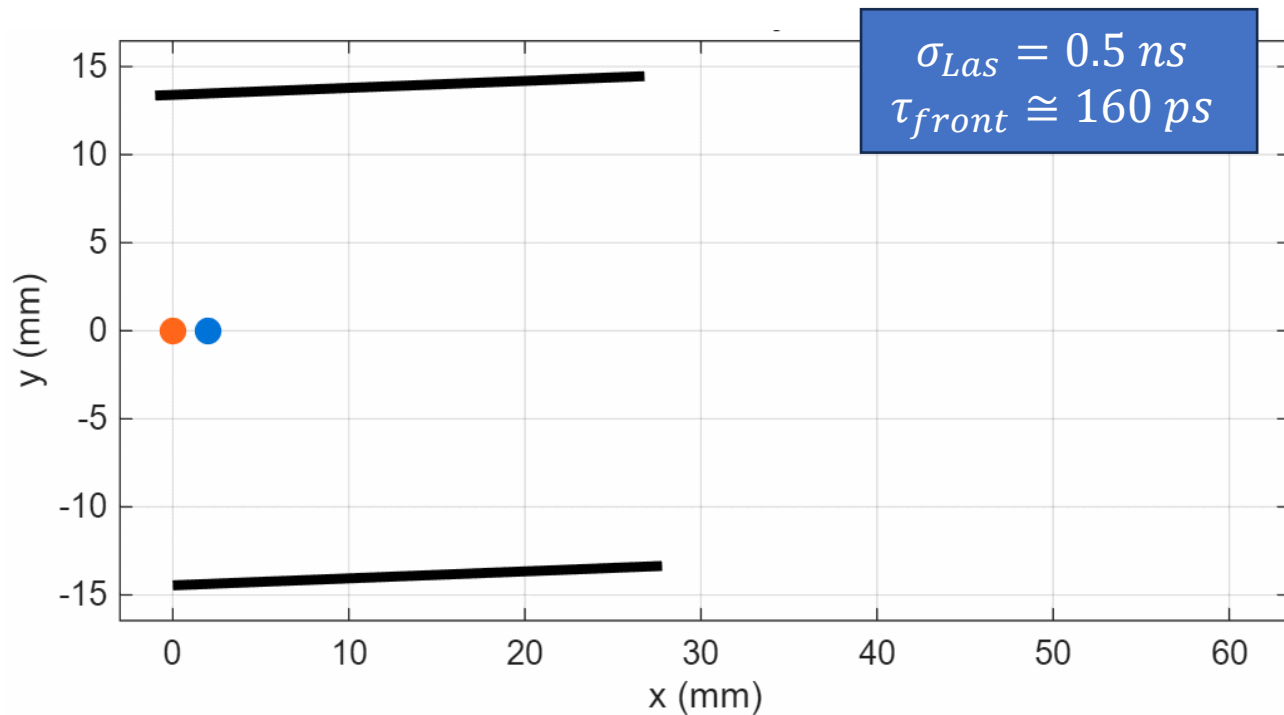


TraceWin model of a particle
distribution out of the Fermilab RFQ

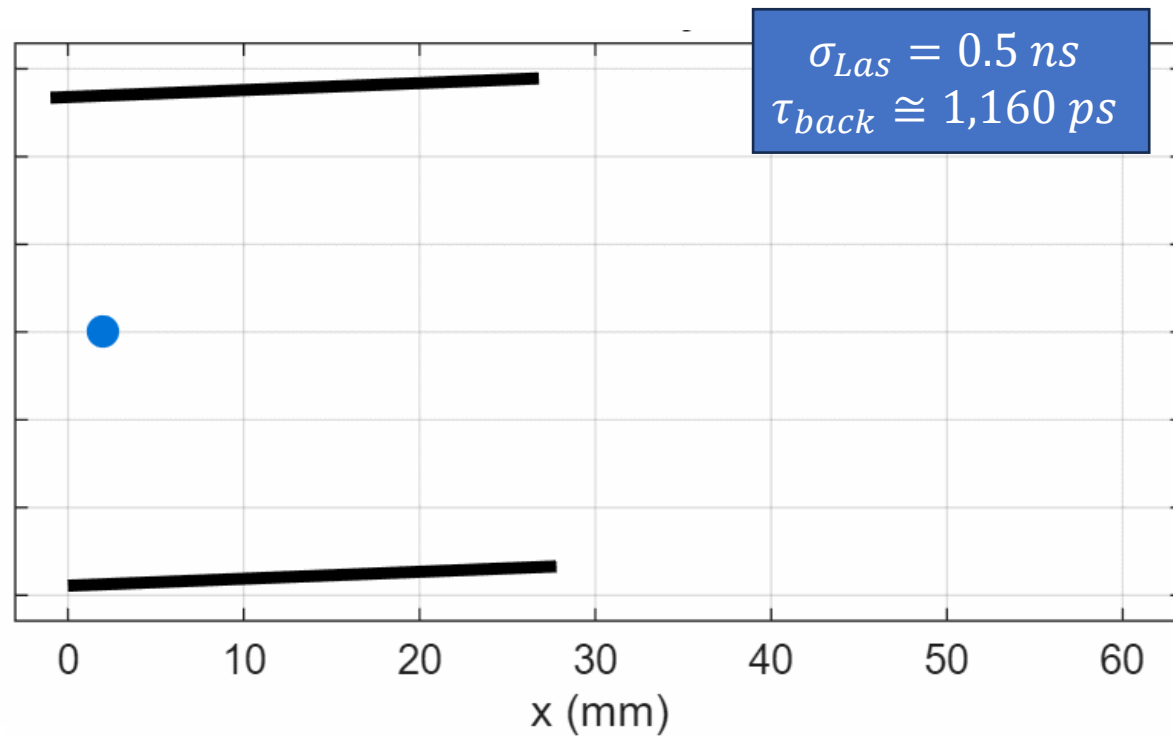
Introducing the Laser Notcher



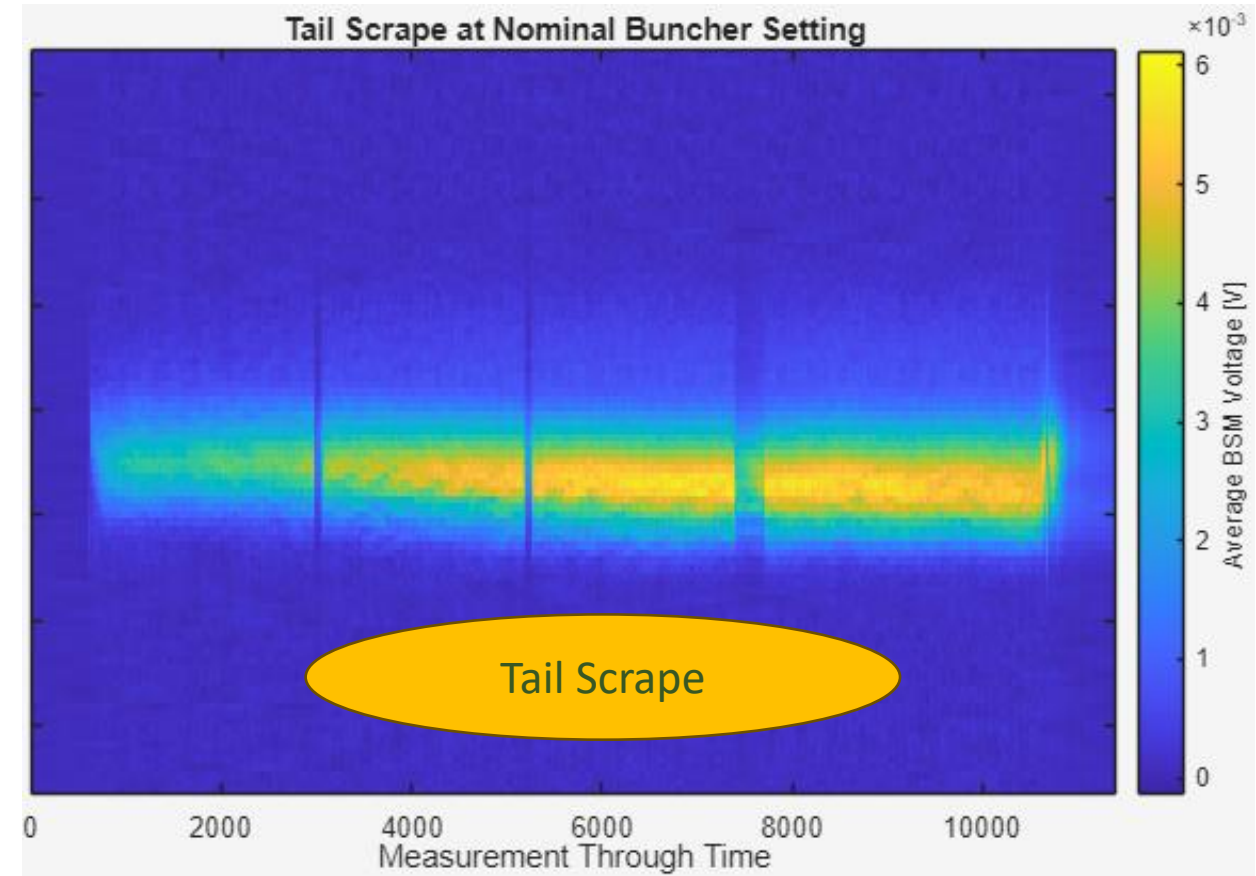
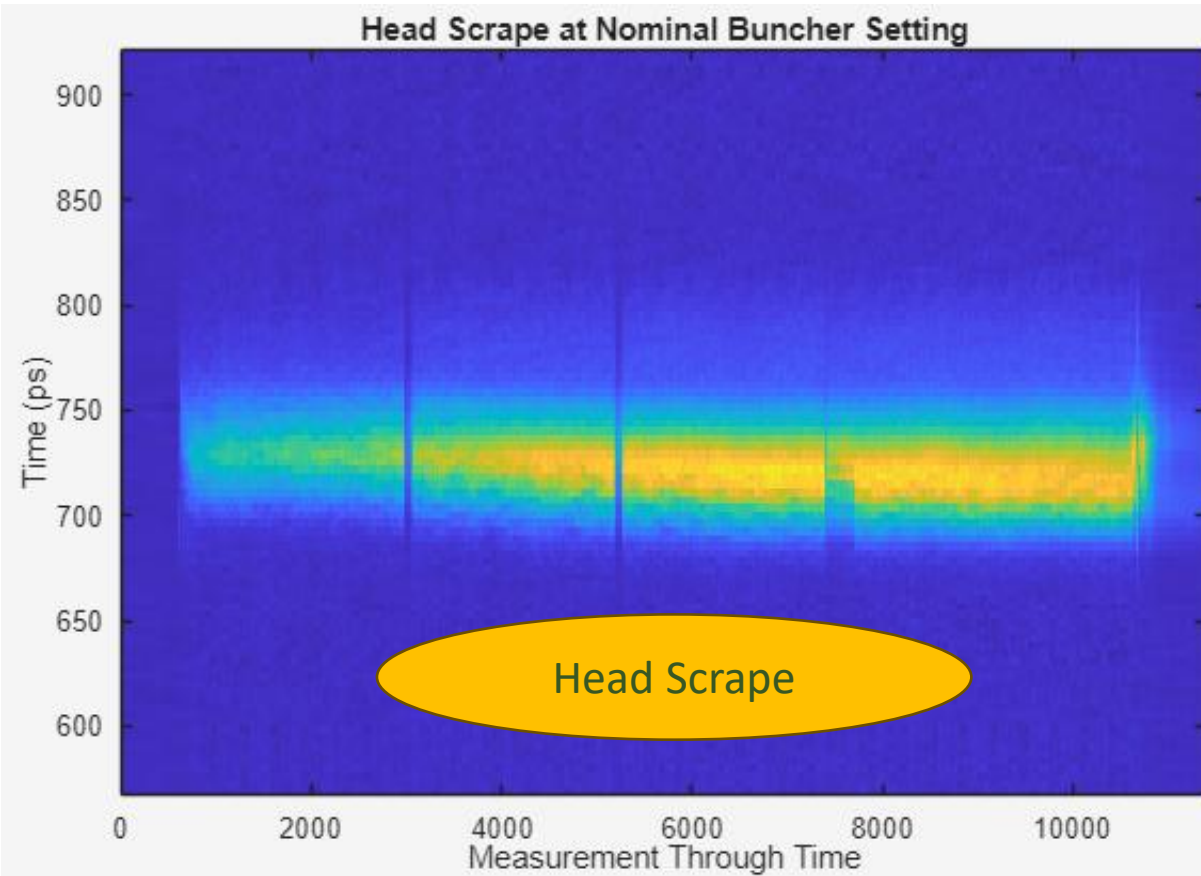
Head Scrape



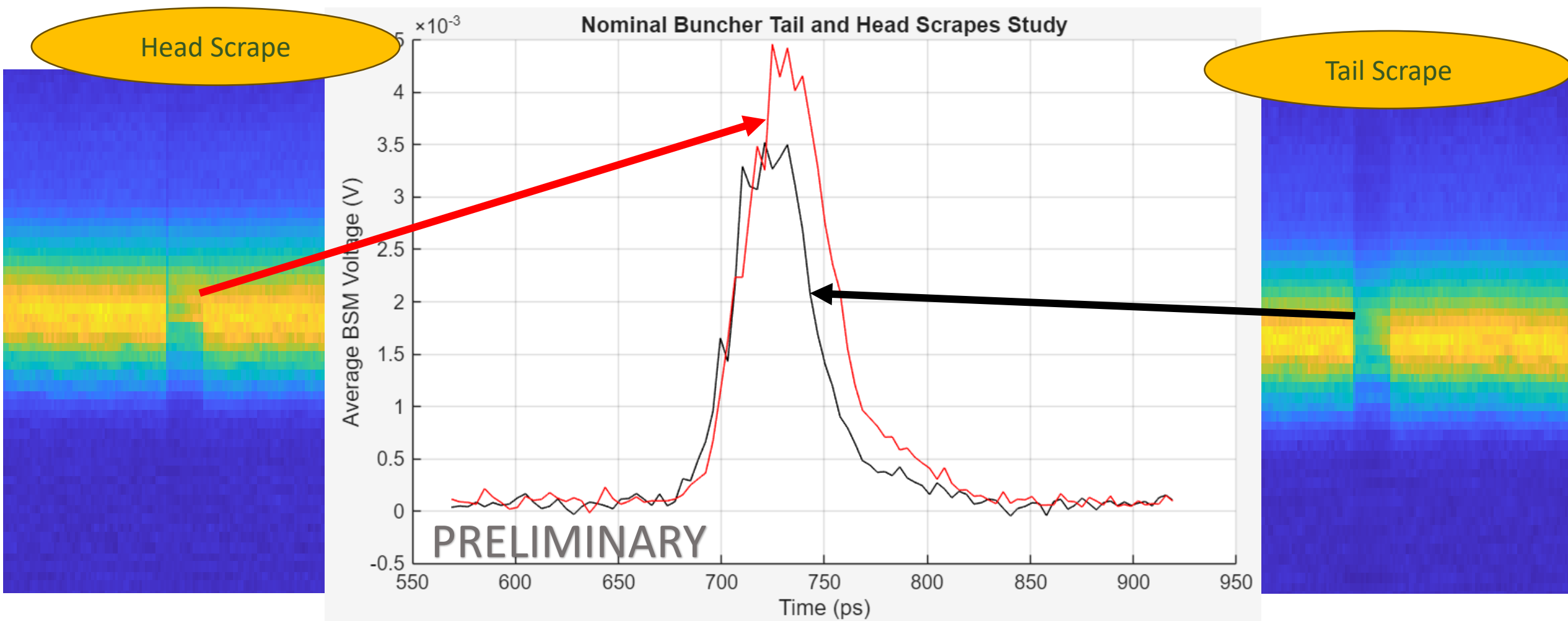
Tail Scrape



Investigating Beam Trend Over Time

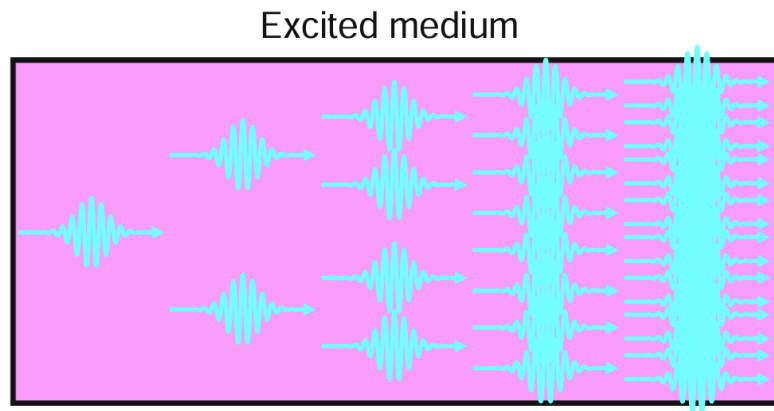


Investigating Beam Trend Over Time

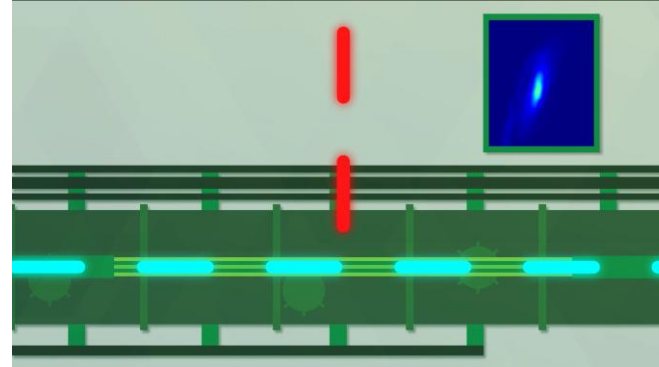


Same Buncher setting, different laser scrapes

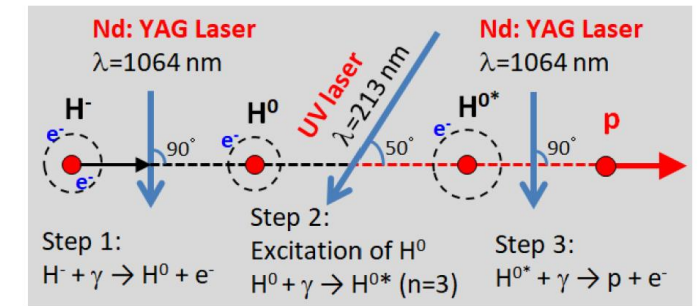
Wrapping up



Laser Wire Instrumentation:



Laser Stripping of H⁻:



Fine Beam Manipulation:

